

NexHire - Smart Recruitment Matching Platform

Final submission document verified against the public main branch.

Item	Final value
Project	NexHire / SmartRecruitmentMatching-2026
Group	Alpha
Repository	https://github.com/Kogulanathan69/SmartRecruitmentMatching-2026-
Visibility	Public
Final branch / commit	main / aac7358
Target framework	.NET 8.0 (net8.0)
Date	11 August 2026

Submission coverage: .NET version, frontend, backend, database, languages, APIs/services, and other tools/libraries are all listed below. No placeholder group name remains.

1. Project Overview

NexHire is a role-based recruitment system for Job Seekers, Employers and Administrators. It supports profile/CV evidence, vacancies, applications, deterministic matching and ranking, status workflows, consent-first contact, notifications and administration.

- Job Seeker: email-OTP registration, profile/evidence, CVs, job search, match explanation, applications and tracking.
- Employer: company details, vacancy management, applicants, ranking/comparison, status updates and contact requests.
- Administrator: controlled platform/user administration through protected admin functions.

2. Solution Architecture

Layer	Location	Responsibility
Presentation / API	src/NexHire.API	Controllers, middleware, DI, auth pipeline, health, OpenAPI and frontend hosting.
Application	src/NexHire.Application	DTOs, validators, interfaces, services, mappings and business rules.
Domain	src/NexHire.Domain	Entities, enums and core recruitment model.
Infrastructure	src/NexHire.Infrastructure	EF Core, repositories, JWT/BCrypt, SMTP, matching calculators and file storage.
Frontend	frontend/	HTML/CSS/JavaScript UI using Fetch against REST APIs.
Tests	tests/	xUnit unit and integration tests with Moq / FluentAssertions.

3. Full Technical Stack

Required item	Technology	Version / use
.NET version	.NET / ASP.NET Core	net8.0; Microsoft package line 8.0.29
Frontend	HTML5, CSS3, JavaScript	Vanilla JS, Fetch API, responsive multi-page UI
Backend	ASP.NET Core Web API	.NET 8 controller-based REST API
Database	Microsoft SQL Server LocalDB	(localdb)\MSSQLLocalDB; NexHireDb
ORM	Entity Framework Core SQL Server	8.0.29; code-first migrations
Languages	C#, JavaScript, HTML, CSS, SQL	Backend, browser UI and scripts
API / services	REST + Swagger/OpenAPI	Swashbuckle 6.6.2; JWT-protected endpoints
Authentication	JWT Bearer	60-minute access token; 7-day refresh token
Password security	BCrypt.Net-Next	4.0.3
Validation	FluentValidation	11.9.2
Mapping	AutoMapper	15.1.3
Email / OTP	SMTP / Gmail	System.Net.Mail; smtp.gmail.com:587 + TLS
File storage	Local file storage	Resume/CV storage service
Testing	xUnit, Moq, FluentAssertions	Unit + integration tests
Other tools	Git, GitHub Actions, Docker, VS, PowerShell	Version control, CI, container build and local development

4. Key Libraries / Packages

Package	Version	Purpose
Microsoft.AspNetCore.Authentication.JwtBearer	8.0.29	JWT auth
Microsoft.EntityFrameworkCore.SqlServer	8.0.29	SQL Server provider
Microsoft.EntityFrameworkCore.Design	8.0.29	Migration tooling
Swashbuckle.AspNetCore	6.6.2	Swagger/OpenAPI
FluentValidation.DependencyInjectionExtensions	11.9.2	Validation
AutoMapper	15.1.3	DTO/model mapping
BCrypt.Net-Next	4.0.3	Password hashing
System.IdentityModel.Tokens.Jwt	8.14.0	JWT token handling

5. Database and Data Model

EF Core migrations create/update NexHireDb. Migration history is in src/NexHire.Infrastructure/Data/Migrations; SQL support scripts are under database/Scripts.

- Development connection: Server=(localdb)
\MSSQLLocalDB;Database=NexHireDb;Trusted_Connection=True;TrustServerCertificate=True;
- Persisted areas include users/tokens, profiles/evidence, resumes, companies, jobs, applications, matching results/rules, contact requests, notifications and status history.
- Final main includes the AddEmailVerificationOtp migration.

6. APIs and Services

Area	Representative endpoints / service
Authentication	/api/Auth/register, verify-otp, resend-otp, login, refresh, forgot/reset/change-password, logout, me
Admin / Users	Protected /api/Admin/... administration
Jobs / Applications	/api/jobs, /api/applications
Profiles / Resumes	/api/jobseekers, /api/resumes
Companies	/api/companies
Matching	/api/matching + matching-rule APIs
Workflow	Application status, contact consent, dashboards, notifications
Operations	/health/live, /health/ready, development /swagger

7. Registration OTP

- 6-digit cryptographic OTP; 10-minute expiry; 60-second resend cooldown; maximum 5 invalid attempts.
- New user stays unverified until OTP succeeds; login is blocked before verification.
- SMTP credentials are runtime configuration and are not committed to GitHub.

8. Security and Reliability

- JWT Bearer authentication and role-based authorization.
- BCrypt password hashing; secure token generation; hashed OTP/token storage.
- Central exception handling plus /health/live and /health/ready operational checks.
- GitHub Actions workflow validates restore, build, unit tests, integration tests and Docker build.

9. Final Validation Evidence

Check	Result
Repository	Public
Final branch / commit	main / aac7358
Restore / final build	PASS / PASS
Unit tests	127 / 127 PASS
Integration tests	4 / 4 PASS
Total automated tests	131 / 131 PASS
Database migration	PASS
/health/live	PASS
/health/ready	PASS
Real email OTP	PASS
Verified login + dashboard/pages	PASS
GitHub CI	PASS

Repository link for Google Form: <https://github.com/Kogulanathan69/SmartRecruitmentMatching-2026->